

SAMEEKSHA MAHESH

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EDUCATION

University of California, Davis

September 2025 - June 2027

Master of Science, Computer Science

GPA: 4.0/4.0

- Coursework: Design & Analysis of Algorithms, Distributed Database Systems, Machine Learning

Dayanand Sagar College of Engineering

August 2020 - June 2024

Bachelor of Engineering, Computer Science & Engineering

GPA: 3.97/4.0

TECHNICAL SKILLS

Programming Languages: Java, Python, C, C++, SQL, TypeScript, JavaScript, Perl

AI/ML: LLM APIs, RAG Pipelines, LangChain, XGBoost, MLflow, TensorFlow, Vector Embeddings, Feast, Hugging Face

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB, Redis, Snowflake, Elasticsearch

Frameworks & Libraries: Spring Boot, React, FastAPI, Node.js, Django, Flask, SQLAlchemy, PyTorch, Apache Kafka, Spark (PySpark), REST APIs

Developer Tools & Cloud: AWS (EC2, S3, EBS), Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions), Git, Databricks, Linux, JUnit, Postman

PROFESSIONAL EXPERIENCE

Health Tequity

July 2026 - Present

AI Engineer Intern

- **Took ownership of the full development cycle** for a feature — design, build, deployment, and operation — across a **TypeScript** frontend and a **Java/Spring Boot** backend, with zero regressions to workflows users already depended on.
- Integrated a third-party service that **validates and standardizes** incoming records against an authoritative reference, and modeled the data against a regulated industry standard.

Perfios Software Solutions Pvt. Ltd.

April 2024 - September 2025

Software Engineer (Financial Technology)

- **Partnered with enterprise clients** to design, build, and roll out a full-stack application from scratch into **production** — a **Java/Spring Boot** backend with role-based access control, OAuth2/JWT authentication, and AES-256 field-level encryption over sensitive customer financial data.
- **Optimized a complex production workflow** — profiling hot paths to locate the real bottleneck, then redesigning caching and asynchronous request handling to cut **REST API response times by 20%** across 5+ integrated services under live customer load.
- Built and operated 8+ containerized services in a **distributed microservices** architecture with **Docker, Kubernetes**, and automated **CI/CD** in two-week **Agile** sprints — **deployment failures down 40%** — under **peer code review** on every change.
- Raised automated document-match accuracy **from 25% to 65%** by analyzing where an automated pipeline was failing against expected results, then rebuilding it over inconsistent real-world data.

Brane Enterprises Pvt. Ltd.

June 2022 - January 2023

Automation Intern

- Wrote 20+ automated **integration and regression tests** in Postman that validated correctness end to end, increasing test coverage by 25%, and integrated them into the **CI** pipeline.

PROJECTS

Real-Time Fraud Detection Platform | github.com/sameeksham09/fraud-feature-store

June 2026

- Engineered a **distributed, scalable** service flagging fraudulent **financial transactions** in real time — **Apache Kafka** ingesting a continuous event stream into **Spark (PySpark)** Structured Streaming, computing derived state into a dual online (**Redis**) and durable offline store so live results stay consistent with those computed in batch.
- Served **sub-second** predictions through a **FastAPI** service behind a version-tracked model registry (**MLflow**), deployed as **5+ independently deployable containerized services** with documented data contracts between stages.

Localbite — Full-Stack Web Application (Team Project) | github.com/localbite-davis/localbite

February 2026

- Built front end, back end, and database on a small **cross-functional team** for a platform serving **1,000+ users** — a **React** client over a **Python/FastAPI** backend on versioned **REST APIs**, with a **PostgreSQL** schema designed to evolve through migrations without breaking live data.
- Broke the feature set into well-defined, shippable components delivered in weekly iterations against user feedback, and implemented **JWT** authentication with role-based access across three user roles.

Agentic Document QA System | github.com/sameeksham09/GenAI-Document-Reader

December 2025

- Developed the system using **AI coding tools** to move faster while **keeping responsibility for the quality of the code** — reviewing and verifying every AI-generated change before integrating it rather than accepting output on trust.
- Designed the **automated benchmark and evaluation harness** that decided **when to trust the AI output and when not to** — a held-out reference set, automated accuracy metrics, and audit logging of every query and source — then drove each change from measured results: **answer accuracy (Token F1) improved from 0.44 to 0.73**.
- Implemented the underlying pipeline in **Python** — hybrid keyword (**BM25**) and semantic retrieval over a vector index with two-stage reranking — orchestrated with **LangChain** and served as a streaming **FastAPI REST API**.